ADDENDUM ON FINITE PRESENTATION: THE THICKENING W' TO W IS OF FINITE PRESENTATION, AND GENERIC-ORBIT POINTS LIE IN W

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Allowed standard prerequisites (editorial).

- The R-invariant thickening decomposition with the covering property (more-groupoids chapter, Lemma lemma-finite-flat-over-almost-dense-subscheme, canonical tag 0ABA; collected on this lane as L22511). cited at statement lines 2753-2754 and proof lines 2781-2787. The base lemma being strengthened: its construction W = coprod W_r with W_r inside Z_{r - 1} without Z_r and its covering give the frame for part (2).
- The finite stratification U = Z_0 superset Z_1 superset ... (more-groupoids chapter, Lemma lemma-finite-stratify, canonical tag 0AB9). cited at proof lines 2764-2766. Provides the stratification whose finite presentation refinement and invariance drive both parts.
- Pullback description of the strata of the finite stratification, of finite presentation when s, t are (divisors chapter, Lemma divisors-lemma-locally-free-rank-r-pullback, canonical tag 05P8). cited at proof lines 2766-2768. Gives that the closed immersions Z_k to U are of finite presentation, from which part (1) follows.
- Chevalley's theorem: images of constructible sets under finitely presented morphisms are constructible (morphisms chapter, Theorem morphisms-theorem-chevalley, canonical tag 054K; kind theorem). cited at proof lines 2775-2776. The images of Z_k to U and Z_{k + 1} to U are locally constructible, the input for the generic point neighbourhood argument.
- A generic point of an irreducible component has a neighbourhood inside a constructible set containing it (properties chapter, Lemma properties-lemma-generic-point-in-constructible, canonical tag 0AAW). cited at proof lines 2777-2780. Produces the open neighbourhoods U_i of the orbit points inside Z_k without Z_{k + 1} set theoretically.

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Source reference. Original lemma 0ABA: sources/stacks-chow/tags.txt, source lines 13013–13013. Cited at statement lines 2753-2754 as the base lemma being strengthened and at proof lines 2781-2787 for the construction W = coprod W_r and the covering used in the stratum matching.

Source reference. Original lemma 0AB9: sources/stacks-chow/tags.txt, source lines 13012–13012. Cited at proof lines 2764-2766: the stratification U = Z_0 superset Z_1 superset ... under finite presentation assumptions.

Source reference. Original lemma 05P8: sources/stacks-chow/tags.txt, source lines 7332–7332. Cited at proof lines 2766-2768: the strata are closed immersions of finite presentation, yielding part (1).

Source reference. Original theorem 054K: sources/stacks-chow/tags.txt, source lines 6643–6643. Cited at proof lines 2775-2776: the images of Z_k to U and Z_{k + 1} to U are locally constructible.

Source reference. Original lemma 0AAW: sources/stacks-chow/tags.txt, source lines 12999–12999. Cited at proof lines 2777-2780: the open neighbourhoods U_i of the generic orbit points inside Z_k without Z_{k + 1}.

Lemma 0.2. *In Lemma 0ABA assume in addition that s and t are of finite presentation. Then*

(1) *the morphism $W' \to W$ is of finite presentation, and*
(2) *if $u \in U$ is a point whose R-orbit consists of generic points of irreducible components of U, then $u \in W$.*

Proof. In this case the stratification $U = Z_0 \supset Z_1 \supset Z_2 \supset \ldots$ of Lemma 0AB9 is given by closed immersions $Z_k \to U$ of finite presentation, see Divisors, Lemma 05P8. Part (1) follows immediately from this as $W' \to W$ is locally given by intersecting the open W by Z_r. To see part (2) let $\{u_1, \ldots, u_n\}$ be the orbit of u. Since the closed subschemes Z_k are R-invariant and $\bigcap Z_k = \emptyset$, we find an k such that $u_i \in Z_k$ and $u_i \notin Z_{k+1}$ for all i. The image of $Z_k \to U$ and $Z_{k+1} \to U$ is locally constructible (Morphisms, Theorem 054K). Since $u_i \in U$ is a generic point of an irreducible component of U, there exists an open neighbourhood U_i of u_i which is contained in $Z_k \setminus Z_{k+1}$ set theoretically (Properties, Lemma 0AAW). In the proof of Lemma 0ABA we have constructed W as a disjoint union $\coprod W_r$ with $W_r \subset Z_{r-1} \setminus Z_r$ such that $U = \bigcup t(s^{-1}(\overline{W_r}))$. As $\{u_1, \ldots, u_n\}$ is an R-orbit we see that $u \in t(s^{-1}(\overline{W_r}))$ implies $u_i \in \overline{W_r}$ for some i which implies $U_i \cap W_r \neq \emptyset$ which implies $r = k$. Thus we conclude that u is in

$$W_{k+1} = Z_k \setminus \Big(Z_{k+1} \cup \bigcup\nolimits_{r \leq k} t(s^{-1}(\overline{W_r})) \Big)$$

as desired. □